

Blended Learning in Higher Education: Definitions, Models, Technologies, Outcomes, and Implementation Barriers

ABSTRACT

Blended learning (BL) integrates physical and virtual learning environments through different combinations of modality, timing, participation, and technological support to create interactive, flexible, and practical educational experiences in higher education. This scoping-oriented review analyzes recent literature to address seven research questions: (1) definitions and terms used to describe BL as an interactive, technology-enhanced environment; (2) models and strategies that blend in-person and online learning; (3) the prioritization of online sessions and their effects on student engagement; (4) combinations of synchronous and asynchronous activities and their impacts; (5) technologies used to support interactivity and learning outcomes; (6) reported outcomes related to learning, engagement, and satisfaction; and (7) barriers educators face in implementing BL and adopting interactive learning practices. Results reveal diverse terminology, including hybrid learning and the flipped classroom; models that emphasize active learning and flexibility; and technologies such as learning management systems and video platforms that enhance engagement and motivation. However, barriers, including technological challenges, a lack of institutional support, and pedagogical limitations, persist. This review contributes to the design of interactive learning environments by consolidating current knowledge on BL definitions, models, technological integration, learning outcomes, and barriers to implementation. Findings inform the development of blended courses that maximize interactivity, engagement, and learning effectiveness in higher education. They provide actionable guidance for instructional design, faculty development, and institutional policy frameworks to support the effective integration of technology-enhanced blended learning environments.

KEYWORDS

Blended learning; Higher education; Hybrid learning; Flipped classroom; Online learning; Educational technology.

Introduction

Context and Definitions

Blended learning in higher education is a pedagogical approach that integrates face-to-face instruction with online components to create interactive, personalized, and flexible learning environments (Ali, 2024b; Alonso et al., 2025; Miranda et al., 2024; Misra et al., 2025a). Although this review focuses on research published from 2015 onward, this conceptualization is consistent with early blended-learning definitions that emphasized the purposeful combination of in-person and digital modalities. As previously observed in the mapping of blended-learning research by Ibarra-Vargas et al. (2023), this integration can be understood across multiple dimensions, including the environments in which learning occurs, the timing of interaction, and the technological means used to connect instructional activities. Accordingly, blended learning is not limited to a synchronous–asynchronous distinction: in practice, different configurations may combine modality, location, participation, and learning pathways, as illustrated by flipped, hybrid, HyFlex, rotational, enriched-virtual, and other flexible designs.

Synchronous and asynchronous arrangements represent one of these possible dimensions (Price & Winchester, 2025; Shah et al., 2024), and their integration with other design dimensions is not merely structural: it enables iterative cycles of preparation, practice, and reflection that deepen self-directed learning, strengthen higher-order thinking, and more effectively support the development of professional competencies (Adigun et al., 2024; L. Luo, 2025). The approach fosters student engagement, satisfaction, and motivation (Han, 2025; Lv & Li, 2024; Sun et al., 2025) while promoting collaboration and peer support (Kyei-Akuoko et al., 2025; Mohamed, 2024; Zhu et al., 2025).

Blended learning has emerged as an integral, dominant model in universities, accelerated by rapid technological advancements and post-pandemic transitions, because these conditions exposed structural limits of entirely in-person instruction and validated technology-enhanced modalities as scalable, resilient pedagogical solutions (Ali & Georgiou, 2024; Kok et al., 2025; Raes et al., 2020; She et al., 2025; Trabelsi et al., 2025). Its constructivist framing emphasizes interactive learning pathways that foster creativity, engagement, and resilience (Adigun et al., 2024; Ang et al., 2024; J. Yang et al., 2024). Despite widespread adoption, its implementation requires institutional support, professional development, and technological infrastructure (Garone et al., 2025; Y. Peng et al., 2025; Sanders & Mukhari, 2024). Faculty perspectives, self-efficacy, workload, and attitudes significantly influence adoption (Ali, 2024b; ElSayad, 2024; McCarthy et al., 2025). Students benefit from

flexibility and personalized experiences, but may face challenges such as low participation, feelings of loneliness, and varied academic performance, depending on the design and discipline (Harper et al., 2024; Pei et al., 2025; She et al., 2025). The effectiveness of blended learning is context-dependent and influenced by motivational, structural, and environmental factors (Abbasi et al., 2025; Yusuf et al., 2024).

Overall, blended learning represents an innovative, digitally mediated educational paradigm that integrates pedagogical and technological elements to enrich learning environments, aligning with interactive learning environments to foster engagement, adaptability, and lifelong learning skills in higher education (Anthony Jnr, 2024; Casado García-Hirschfeld et al., 2025; Dieu, 2025). In this review, these developments are interpreted through constructivism, the Community of Inquiry framework, and the Technological Pedagogical Content Knowledge model, which together explain how interaction, cognitive engagement, and technology–pedagogy alignment shape blended learning in higher education. The following section outlines the theoretical frameworks that guide the interpretation of the evidence synthesized in this review.

Theoretical Foundations

Expanding upon the concepts briefly introduced in the previous section, this section provides a detailed explanation of the three primary theoretical frameworks used in blended learning research. Blended learning in higher education integrates constructivism, the Community of Inquiry (CoI) framework, and the [Technological Pedagogical Content Knowledge \(TPACK\) model across configurations that combine physical and virtual environments, temporal arrangements, participation modes, and technological resources](#) to foster active, student-centered learning, and in this review these three frameworks provide the conceptual lens for interpreting definitions, models, technological integration, learning outcomes, and implementation barriers (Dillenbourg, 2013; Misra et al., 2025b, 2025a; Sun et al., 2025).

[These frameworks were selected because they provide complementary perspectives on the dimensions examined in the review: constructivism explains active and collaborative knowledge construction; CoI explains how cognitive, social, and teaching presence shape interaction and engagement; and TPACK explains the alignment among technology, pedagogy, and disciplinary content. Accordingly, they support the interpretation of pedagogical models and temporal arrangements in RQ2–RQ4, technological integration in RQ5, learning and engagement outcomes in RQ6, and implementation barriers in RQ7. They are used as interpretive lenses for the synthesized evidence rather than as eligibility criteria or as a formal coding scheme. From a constructivist perspective, learning](#)

is understood as an active process of knowledge construction in which social interaction and guided participation play a central role ([Vygotskij & John-Steiner, 1979](#)).

Flipped classrooms exemplify constructivist principles by enabling students to engage with online materials before class and participate in collaborative, problem-based activities during class, thereby enhancing higher-order thinking, self-directed learning, and motivation across various disciplines, including medicine, STEM ([science, technology, engineering, and mathematics](#)), business, and language education (Dečman et al., 2025; Mortensen & Nicholson, 2015). Constructivism explains these effects by emphasizing that learners reorganize prior knowledge through meaningful, socially mediated tasks. In blended formats, asynchronous preparation activates foundational schemas, while synchronous interaction provides the negotiation and feedback necessary for conceptual change. This mechanism clarifies why models with strong pre-class preparation and collaborative in-class work consistently yield deeper cognitive engagement.

The Community of Inquiry (CoI) framework was originally developed to explain educational experience in computer-mediated learning through the interaction of cognitive, social, and teaching presence ([Garrison et al., 1999](#)). Its later application to blended learning extends these three forms of presence across face-to-face and online activities. The CoI framework is operationalized through blended learning, which promotes cognitive, social, and teaching presence through multi-directional interaction, reflective discussions, peer-supported learning, and robot-mediated or videoconference communication, thereby enhancing social presence and authentic participation (Gleason & Greenhow, 2017; Kreijns et al., 2003; Miranda et al., 2024; Ramírez-Mera et al., 2025). The CoI framework provides a functional explanation for these outcomes. Cognitive presence develops when asynchronous materials promote inquiry, while teaching presence depends on how instructors structure synchronous and asynchronous interactions. Social presence predicts participation and satisfaction, explaining why poorly designed online phases weaken engagement even when the overall blended structure appears sound. Teaching presence is reinforced by design models that emphasize clear communication, early planning, and the appropriate use of technology (J. Lee et al., 2017; Miranda et al., 2024).

The Technological Pedagogical Content Knowledge (TPACK) framework conceptualizes effective technology integration as the interaction among technological knowledge, pedagogical knowledge, and content knowledge, together with their intersecting forms of knowledge ([Mishra & Koehler, 2006](#)). This perspective is particularly relevant to blended learning because effective course design requires technological choices to be aligned

simultaneously with instructional strategies and disciplinary content. Technological integration, as in the TPACK model, is evident through Learning Management Systems (LMS), mobile apps, Massive Open Online Courses (MOOCs), hypermedia, digital storytelling, and artificial intelligence, supporting flexible, personalized, and competency-based learning (Giang et al., 2025; Glahn et al., 2015; Peres et al., 2017). TPACK also clarifies why blended learning outcomes vary across institutional and disciplinary contexts. When one domain—technological, pedagogical, or content—is misaligned, instructional coherence weakens and engagement declines. This framework therefore accounts for many reported implementation barriers, including ineffective tool selection, inconsistent instructional design, and limited digital pedagogical expertise.

Blended learning environments foster autonomy-supportive, interactive, and resilient learning spaces (Boeren, 2016), while hybrid designs enhance problem-solving skills, critical thinking, digital competencies, and graduate employability (R. Ismail et al., 2016; Thai et al., 2017). Institutional support, including policies, technological infrastructure, and faculty development, is crucial for effective implementation (Y. Peng et al., 2025; Protsiv & Atkins, 2016). Systematic reviews identify instructional design, interactions, and learner outcomes as key trends in blended learning (Pima et al., 2018). Despite challenges such as time demands and infrastructure (Kumpu et al., 2016; Protsiv & Atkins, 2016), blended learning operationalizes these theories to create sustainable, technologically integrated, socially rich educational practices that enhance engagement, performance, and satisfaction in higher education contexts (Ghai et al., 2025; Khodabandelou et al., 2016).

Taken together, constructivism, the CoI framework, and the TPACK model therefore frame blended learning not merely as a delivery modality, but as an interactive learning environment in which pedagogy, technology, and social presence are intentionally aligned to support engagement, autonomy, and higher-order learning in higher education. These theoretical frameworks also illuminate the patterns identified in the remainder of the review. Constructivism clarifies the role of active and collaborative learning; CoI explains how interaction quality shapes engagement and satisfaction; and TPACK accounts for the technological–pedagogical alignment required for effective implementation. Using these lenses strengthens the analytical interpretation of the findings across all research questions.

Models and Temporal Structures in Blended Learning

This section synthesizes how studies operationalize blended learning through models and temporal structures. The following subsections group recurring patterns to reduce terminological inconsistencies and clarify how

different designs shape interactive learning processes.

Terminology and Conceptual Inconsistencies

Conceptualization in blended learning remains fragmented, with diverse and inconsistently used terms such as flipped learning, hybrid teaching, flexible learning, and agile blended learning across disciplines (Baig & Yadegaridehkordi, 2023; Gudoniene et al., 2025; Halder et al., 2024; Kolstad et al., 2024). This definitional ambiguity constrains the development of pedagogical frameworks, technological integration, and quality assurance interventions, ultimately generating inconsistencies in course design and limiting institutions' capacity to evaluate whether blended learning implementations genuinely enhance engagement and learning quality (Benson et al., 2024; X. Li et al., 2024; Ramsey, 2024). Moreover, inconsistent terminology limits cross-disciplinary synthesis and hinders the advancement of interactive learning theories (Ishmuradova et al., 2024; McCarthy & Palmer, 2023). Clarifying how blended learning is defined and described is necessary to support effective instructional design, enhance student motivation and skill acquisition, and ensure its transformative implementation within higher education (Cao et al., 2025; Castro-Rodríguez et al., 2021; C. Wang et al., 2022), ultimately aligning with the goals of creating engaging, effective, and healthy educational settings.

Diversity of Pedagogical Models

Models remain fragmented, with diverse approaches such as flipped classrooms, hybrid teaching, agile blended learning, and enriched virtual models used without conceptual clarity (Baig & Yadegaridehkordi, 2023; Haftador et al., 2023; Halder et al., 2024; Naing et al., 2023). This inconsistency constrains pedagogical frameworks, technological integration, and instructional design, which are essential for fostering interactive, engaging, and effective learning environments (Adigun et al., 2024; Benson et al., 2024; R.-Z. Luo & Zhou, 2024). Studies highlight positive impacts on motivation, autonomy, and skill acquisition (Alonso et al., 2025; Castro-Rodríguez et al., 2021; Niu et al., 2023), yet challenges such as technological barriers, teacher competencies, and assessment complexities persist (Ashraf et al., 2022; Gudoniene et al., 2025). There is a need to identify and analyze models and strategies that maximize student-centered learning, collaboration, and well-being (Halder et al., 2024; Ramsey, 2024), ensuring blended learning meaningfully transforms higher education into interactive, ethical, and personalized environments aligned with lifelong learning goals (Cao et al., 2025; McCarthy & Palmer, 2023; C. Wang et al., 2022).

Prioritization of Online versus In-Person Components

Models vary in their prioritization of online sessions, ranging from flipped classrooms with pre-class video engagement to hybrid and enriched virtual models that distribute online and in-person components differently (Baig & Yadegaridehkordi, 2023; Haftador et al., 2023; Suri & Zaim, 2023; Uwimana & Mukama, 2023). This variation affects student motivation, self-regulation, and interactivity (X. Li et al., 2024; Niu et al., 2023; Ramsey, 2024). Studies highlight benefits for knowledge, skills, and autonomy (Alonso et al., 2025; Castro-Rodríguez et al., 2021; Niu et al., 2023), yet technological barriers, insufficient pedagogical frameworks, and inequitable access persist (Ashraf et al., 2022; Castro-Rodríguez et al., 2021; Lara-Lara et al., 2023). There is a need to examine how the emphasis on online sessions influences engagement, critical thinking, and inclusion to ensure that blended learning fosters interactive, practical, and ethically grounded environments (Lara-Lara et al., 2023; McCarthy & Palmer, 2023; C. Wang et al., 2022). Such clarity supports innovative design, technological integration, and student-centered learning, all of which are aligned with lifelong learning goals (Cao et al., 2025; R.-Z. Luo & Zhou, 2024).

Several studies reveal varied models with inconsistent balances, ranging from flipped classrooms that prioritize asynchronous pre-class preparation with synchronous in-class interaction (Baig & Yadegaridehkordi, 2023; W. Kong et al., 2024; Naing et al., 2023), to hybrid and enriched virtual approaches that integrate components differently (Gudoniene et al., 2025; Haftador et al., 2023). This variation affects motivation, self-regulated learning, cognitive engagement, and overall outcomes (X. Li et al., 2024; Niu et al., 2023; Ramsey, 2024). While evidence highlights benefits for skill acquisition, autonomy, and critical thinking (Alonso et al., 2023, 2025; Castro-Rodríguez et al., 2021), challenges persist, including technological barriers, insufficient pedagogical frameworks, and inequitable access (Ashraf et al., 2022; Castro-Rodríguez et al., 2021; Gudoniene et al., 2025). There is a need to understand how synchronous and asynchronous combinations impact learning to inform innovative, ethical, and student-centered designs (Lara-Lara et al., 2023; McCarthy & Palmer, 2023; C. Wang et al., 2022), ensuring that blended learning environments foster engaging, effective, and inclusive educational experiences aligned with lifelong learning goals (Cao et al., 2025; R.-Z. Luo & Zhou, 2024).

Technological Integration in Blended Learning

Challenges such as technological barriers, communication difficulties, and insufficient pedagogical frameworks persist (Ashraf et al., 2022; Gudoniene et al., 2025). While evidence shows technologies improve motivation, autonomy, and cognitive engagement (Alonso et al., 2023; Castro-Rodríguez et al., 2021; Niu et al.,

2023), their specific contributions to interactive learning and outcomes remain underexplored (Hinojo Lucena et al., 2019; McCarthy & Palmer, 2023). There is a necessity to clarify which technologies foster ethical, practical, and student-centered blended learning (R.-Z. Luo & Zhou, 2024; C. Wang et al., 2022), ensuring their integration transforms educational environments into interactive, engaging, and personalized experiences aligned with lifelong learning goals (Cao et al., 2025; Ishmuradova et al., 2024).

Learning Outcomes Associated with Blended Learning

Diverse models lack a systematic synthesis of their impacts on interactive design (Ishmuradova et al., 2024; McCarthy & Palmer, 2023). There is a need to understand how learning, engagement, and satisfaction outcomes inform practical, ethical, and student-centered blended learning designs (Bashir & Lapshun, 2025; C. Wang et al., 2022), ensuring environments that are engaging, personalized, and aligned with lifelong learning goals (Cao et al., 2025; R.-Z. Luo & Zhou, 2024).

Barriers to Blended Learning Implementation

Educators face persistent barriers to adopting interactive learning practices. Studies identify technological problems, communication difficulties, time management issues, limited institutional support, and insufficient pedagogical frameworks as key challenges (Ashraf et al., 2022; Gudoniene et al., 2025; Halder et al., 2024; X. Li et al., 2024). Additionally, technical incompetence, the lack of comprehensive models to address multi-level barriers, and limited access to resources constrain effective implementation (Castro-Rodríguez et al., 2021; Muhuro & Kang'ethe, 2021; Sareen & Mandal, 2024). Educator preparedness, instructional styles, and technological competence significantly influence student satisfaction and acceptance (Cao et al., 2025; Neville et al., 2021), whereas fragmented research and inadequate policy guidelines hinder integration (Lundin et al., 2018; Muhuro & Kang'ethe, 2021). Clarifying these barriers is necessary to design ethical, flexible, and student-centered interactive environments (Bashir & Lapshun, 2025; R.-Z. Luo & Zhou, 2024), which foster creativity, self-regulated learning, and well-being (Ramsey, 2024; C. Wang et al., 2022). Addressing these issues supports sustainable, engaging, and transformative blended learning in higher education (Cao et al., 2025; Istenič, 2024).

Research Questions and Purpose

Despite the breadth of existing theoretical foundations, at the time this paper was written Scopus indexed 78 reviews on blended learning in higher education published since 2015, yet persistent gaps in understanding and

practice remain. Although adoption has expanded, several critical aspects continue to be underexplored and require systematic investigation, warranting a new review to consolidate evidence and clarify the field's conceptual and practical inconsistencies. In response to these gaps, this paper addresses the following research questions:

- RQ1) How is blended learning defined in higher education, and what terms are used to describe these interactive, technology-enhanced environments?
- RQ2) What models or strategies blend face-to-face and online learning in higher education to create interactive and practical experiences?
- RQ3) To what extent do blended learning models prioritize online sessions, and how does this affect student engagement and interactivity?
- RQ4) How do blended learning environments combine synchronous and asynchronous activities, and what impacts do these balances have on engagement and outcomes?
- RQ5) What technologies are used in higher education blended learning, and how do they support interactivity and learning outcomes?
- RQ6) What student learning, engagement, and satisfaction outcomes are reported in blended learning, and how do these inform effective interactive design?
- RQ7) What barriers do educators face in implementing blended learning, and how do these affect the adoption of interactive learning practices?

This paper aims to [conduct a scoping-oriented review of blended learning in higher education, mapping and synthesizing its definitions, models, technologies, reported outcomes, and implementation barriers as components of interactive learning environments](#). Despite its widespread adoption, blended learning remains conceptually fragmented, with varied terminology and inconsistent frameworks that limit pedagogical design and technological integration. By answering the research questions, this review examines how blended learning is defined, the models and strategies employed, the prioritization of online sessions, the combination of synchronous and asynchronous activities, the technologies used, reported learning and engagement outcomes, and barriers faced by educators. Addressing these gaps will inform the design of flexible, ethical, and student-centered interactive environments that foster motivation, engagement, and lifelong learning skills aligned with higher education goals.

The remainder of this article is organized as follows: the next section presents the methodology used, followed

by the results and the discussion, and concludes with the main findings and implications.

Materials and Methods

This section outlines and critically discusses the [scoping-oriented review workflow](#) used to map and synthesize the literature across the seven research questions. The workflow draws on established approaches to structured literature analysis (Aria & Cuccurullo, 2017; Donthu et al., 2021; Page et al., 2021), while PRISMA guidance is used specifically to support transparent reporting of the identification, screening, eligibility, and inclusion of records rather than to characterize the study as a conventional systematic literature review. The employed method is structured into four key steps:

1. Study design.
2. Data collection and preparation.
3. Data analysis.
4. Data interpretation and synthesis.

This structured approach [supports a transparent and structured synthesis](#) of the literature, enabling a broad understanding of the field of study across the seven research questions.

Study Design

Table 1 outlines the study parameters, following the guidelines proposed by Kitchenham (Kitchenham, 2004). **Fig. 1** illustrates the overall number of documents retrieved from Scopus [using the PRISMA flow structure](#) to transparently report document identification, screening, eligibility, and inclusion. These steps reflect a [scoping-oriented design](#) by emphasizing literature mapping, iterative search refinement, and broad inclusion to capture the range of definitions, models, technologies, outcomes, and barriers in blended learning.

Table 1. Parameters of the Study.

Parameter	Value
Database	Scopus
Years of analysis	10 years
Data retrieval	May 15, 2025. However, it is essential to note that the authors have been working on early versions of this paper since 2020.
Search string	It is derived using an iterative construction method, which will be elaborated upon in the subsequent section.
Inclusion criteria	- Documents published from 2015 onward. - Documents classified as article, conference paper, book chapter, review, book, letter, short survey. - Documents address at least one of the seven research questions
Exclusion criteria	- Documents published <i>before</i> 2015. - Documents classified as data paper, editorial, erratum, note, retracted. - Documents not addressing any of the seven research questions or <i>not selected after the 100 most recent relevant documents had been retained for the corresponding research question.</i>
Quality assessment	None

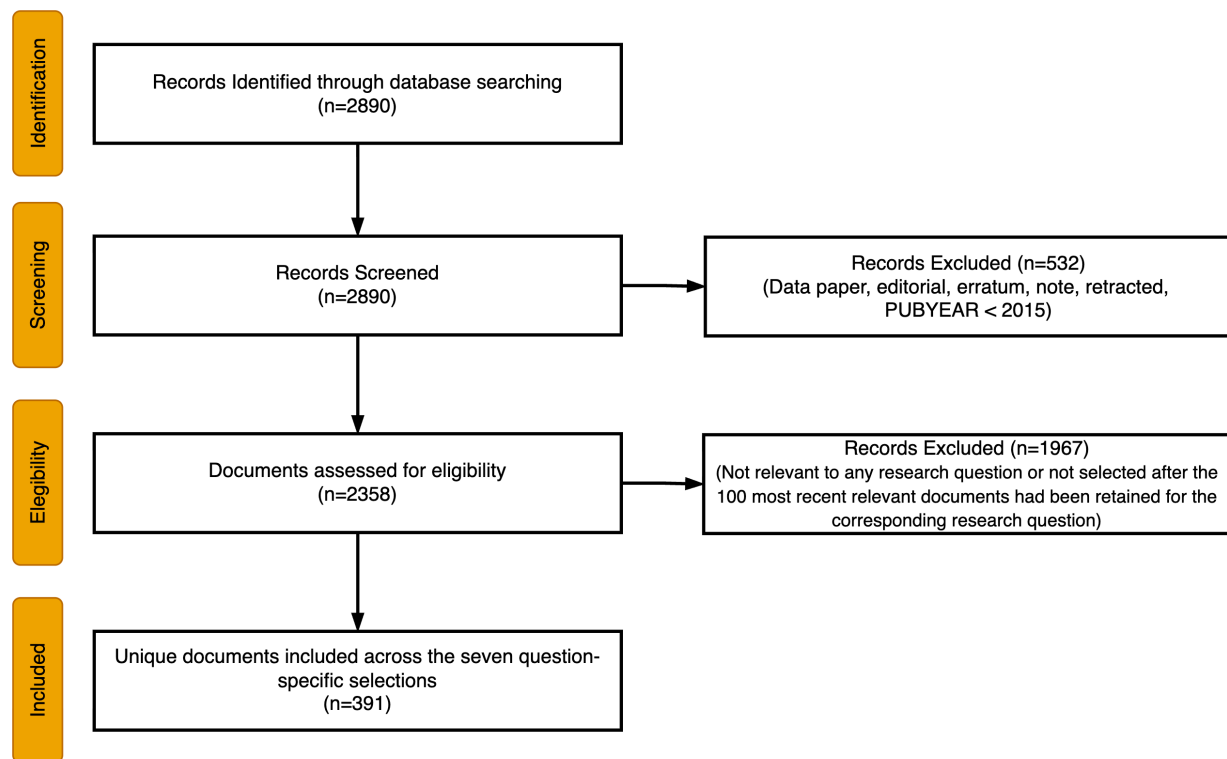


Figure 1. Study-selection flow diagram based on the PRISMA reporting structure. Source: The Authors.

The search string used is:

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TITLE( ( "blended learn*" OR "hybrid learn*" OR "mixed-mode learn*" OR "mixed mode
learn*" OR "blended education" OR "hybrid education" OR "blended course*" OR "hybrid
course*" OR "mixed-mode instruction" OR "mixed mode instruction" OR "technology-
mediated instruction" OR "technology mediated instruction" OR "web-enhanced
instruction" OR "web enhanced instruction" OR "flipped classroom*" OR "flipped
learn*" OR "inverted learn*" ) AND ( "high* education" OR universit* OR "post-
secondary education" OR "postsecondary education" OR college* OR "tertiary education"

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OR "undergraduate" OR "graduate" OR "postgraduate" ))
OR (
  AUTHKEY ( ( "blended learn*" OR "hybrid learn*" OR "mixed-mode learn*" OR "mixed mode
    learn*" OR "blended education" OR "hybrid education" OR "blended course*" OR
    "hybrid course*" OR "mixed-mode instruction" OR "mixed mode instruction" OR
    "technology-mediated instruction" OR "technology mediated instruction" OR "web-
    enhanced instruction" OR "web enhanced instruction" OR "flipped classroom*" OR
    "flipped learn*" OR "inverted learn*" ) AND ( "high* education" OR universit* OR
    "post-secondary education" OR "postsecondary education" OR college* OR "tertiary
    education" OR "undergraduate" OR "graduate" OR "postgraduate" ) )
  AND ABS ( ( "blended learn*" OR "hybrid learn*" OR "mixed-mode learn*" OR "mixed mode
    learn*" OR "blended education" OR "hybrid education" OR "blended course*" OR
    "hybrid course*" OR "mixed-mode instruction" OR "mixed mode instruction" OR
    "technology-mediated instruction" OR "technology mediated instruction" OR "web-
    enhanced instruction" OR "web enhanced instruction" OR "flipped classroom*" OR
    "flipped learn*" OR "inverted learn*" ) AND ( "high* education" OR universit* OR
    "post-secondary education" OR "postsecondary education" OR college* OR "tertiary
    education" OR "undergraduate" OR "graduate" OR "postgraduate" ) ) )

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The search string employed in this study was developed in two parts. The first part aimed to retrieve all documents containing the term "blended learning" or any of its synonyms. The second part focused on retrieving documents in which "blended learning" (or any of its synonyms) appeared simultaneously in both the abstract and the author keywords. Initially, the search string included only the term "blended learning". Additional terms were identified by reviewing the titles, abstracts, and keywords of the retrieved documents. These newly discovered synonyms for blended learning were subsequently incorporated into the search string, and the search and analysis processes were repeated. This iterative procedure continued until no further synonyms were identified.

Data collection and preparation

The designed search string recovered 2890 documents for analysis from Scopus (see **Fig. 1**). In the screening phase, 532 papers were excluded based on metadata inclusion and exclusion criteria (year of publication and document type). As a result, only documents published from 2015 onward were considered. In the eligibility phase, each document was evaluated to determine if it could be used to answer one or more research questions. [For each research question, eligible documents were examined in chronological order, beginning with the most recent.](#) For each research question, only the first 100 most recent documents deemed pertinent were included in the analysis. [The 100-document threshold was therefore applied independently to each research question rather than to the review as a whole, and a document could contribute to more than one research question.](#) This threshold was adopted to foreground the most recent developments in blended learning, given the rapid evolution of technologies and

pedagogical models in the post-2015 period. We acknowledge that this strategy may exclude earlier but influential studies; however, it aligns with the review's goal of providing an up-to-date synthesis of recent evidence. As a result, the documents used to address each question may differ, as selection was based on each document's relevance to its corresponding question. Across the seven question-specific selections, the resulting union comprised 391 unique documents included in the scoping-oriented review. The search relied exclusively on Scopus because it provides broad, multidisciplinary coverage of peer-reviewed research in higher education. This choice may have excluded studies available only in other databases. All bibliographic data from Scopus was downloaded in CSV format. This dataset included document titles, abstracts, author keywords, index keywords, authors, affiliations, source titles, and bibliographies.

Quality assessment

Given the conceptual and mapping-oriented nature of this scoping-oriented review, which aims to synthesize definitions, models, technologies, and implementation barriers of blended learning in higher education, no formal quality assessment of the included studies was conducted, because the study prioritizes breadth of evidence and conceptual mapping over appraisal of individual study rigor. As a result, the evidence synthesized in this review reflects studies that vary widely in design and methodological rigor. This diversity reflects the mapping-oriented purpose of this study, which aims to map and conceptualize the literature rather than assess the quality of individual studies. Consequently, the patterns identified in this review should be interpreted as indicative of trends across a heterogeneous body of studies rather than as precise estimates of effect.

Data analysis

The abstracts were analyzed to answer the corresponding research question. The documents retained for each research question were then examined in greater detail, including full-text analysis of the most relevant studies. The analysis focused on the information directly relevant to each research question, including definitions and terminology, pedagogical models and temporal structures, technologies, reported learning outcomes, and implementation barriers, as applicable. No separate formal coding scheme or codebook was used; instead, the evidence was organized by research question and grouped according to recurring concepts and patterns identified across the studies. Convergences, differences, and recurring findings were then narratively synthesized to produce the results reported for each research question. The Discussion subsequently interprets these synthesized findings

through the theoretical frameworks presented earlier in the manuscript.

Results

The following section presents the results derived from the empirical evidence across the reviewed studies. Interpretive connections to the theoretical frameworks are addressed separately in the Discussion section to maintain a clear distinction between evidence and analytical interpretation. **Table 2** provides a concise synthesis of the main findings across the seven research questions, summarizing the definitions, blended-learning models, online and face-to-face configurations, technologies, reported outcomes, and implementation barriers identified in the reviewed literature.

The following findings reflect the empirical patterns identified across the reviewed literature and, taken together, reveal the structural tendencies, recurring design decisions, and contextual constraints that shape contemporary blended learning practices in higher education. These tendencies align closely with the theoretical foundations presented earlier. Differences in engagement and interaction reflect the dynamics of CoI; the effectiveness of active models is consistent with constructivist principles; and many institutional challenges correspond to TPACK misalignments. This grounding ensures coherence between the theoretical and empirical components of the review.

Table 2. Synthesis of the main findings across the research questions.

Dimension	Main synthesized findings
Definitions and terminology (RQ1)	Blended learning is generally conceptualized as an integrative, flexible, learner-centered approach combining physical and virtual learning environments. The literature uses related terms and configurations including hybrid learning, HyFlex, phygital learning, blended MOOCs, polysynchronous learning, mobile blended learning, and flexible learning.
Models and strategies (RQ2)	The most recurrent models include flipped classrooms, hybrid and HyFlex designs, phygital learning, project-based and problem-based blended learning, MOOC- and SPOC-supported models, dialogic and sequential approaches, and technology-supported active-learning designs.
Online versus face-to-face configuration (RQ3)	Online components commonly support preparation, flexible access, self-paced learning, and content delivery, while face-to-face or synchronous components are primarily used for interaction, application, feedback, problem solving, and collaborative learning. Greater online emphasis is beneficial only when supported by structured interaction, assessment, and pedagogical scaffolding.
Synchronous–asynchronous configuration (RQ4)	A recurrent pattern combines asynchronous pre-class preparation with synchronous interactive activities. Evidence indicates that learning effectiveness depends more on coherent integration between these components than on the proportion of time allocated to either modality.
Technologies (RQ5)	Frequently reported technologies include LMS platforms, instructional videos, screencasts, podcasts, interactive quizzes, AI tools and adaptive systems, ChatGPT, mobile applications and BYOD, MOOCs and SPOCs, augmented reality, digital storytelling, and immersive or problem-based platforms.
Learning, engagement, and satisfaction outcomes (RQ6)	Reported benefits include improvements in critical thinking, problem solving, conceptual understanding, self-regulation, motivation, participation, confidence, collaboration, professional competencies, and satisfaction. However, outcomes are strongly dependent on instructional design, teacher presence, feedback, technological readiness, and student preparation.
Implementation barriers (RQ7)	The principal barriers concern technological infrastructure and connectivity, faculty workload and preparation time, limited institutional support and training, pedagogical and TPACK limitations, resistance to change, student readiness and digital divides, organizational policies, class size, disciplinary traditions, and broader contextual inequalities.

RQ1) How is blended learning defined in higher education, and what terms are used to describe these interactive, technology-enhanced environments?

Responding to RQ1, blended learning in higher education is predominantly defined as an approach that combines traditional face-to-face instruction with online learning components, offering flexibility, convenience, and enhanced learning quality (McCarthy et al., 2025; Misra et al., 2025a; Sun et al., 2025). This integrative pedagogical model is conceptualized as delivering “the best of both worlds,” facilitating access to learning resources, interactivity, cooperation, motivation, and student control over learning (Ghai et al., 2025; Miranda et al., 2024). It is widely framed as a learner-centered, constructivist, and reflective strategy that enables students to engage in cognitive, metacognitive, and social learning processes (Reid & Ewing, 2018; Tuomainen, 2017) and is regarded as essential for personalizing educational experiences (McCarthy et al., 2025). During the COVID-19 pandemic,

blended learning emerged as a predominant instructional mode, transitioning from emergency remote teaching to sustainable practices at activity, course, and programme levels (Lin & Lam, 2025; Yaremko et al., 2025). These definitional trends derive from the empirical analysis of the reviewed studies rather than from the conceptual background outlined in the Introduction, underscoring that practitioners conceptualize blended learning pragmatically—guided by observed effectiveness and contextual demands—rather than by strict theoretical formalizations.

Regarding the terminology associated with blended learning, the reviewed studies identify several models and strategies used to characterize its different configurations. The literature consistently identifies flipped learning or flipped classroom as a core blended learning model, inverting traditional lecture-homework structures by placing knowledge acquisition online before class and using in-person time for active, problem-based, collaborative learning (Dečman et al., 2025; Karthikeyan et al., 2024; J. Lee et al., 2017; Sailer, 2025). This approach supports self-directed learning, higher-order thinking, and aligns with heutagogical paradigms (Green & Schlairet, 2017). Variants include the inverted classroom, where factual learning occurs outside of class, and the flipped classroom, which integrates e-learning before class with teacher interaction during class to foster autonomy and cooperation (Chiang & Wang, 2015; Mortensen & Nicholson, 2015).

Terms such as hybrid learning describe models that blend synchronous online and offline modalities, promoting flexible, multimodal, and equitable learning spaces that blur physical and digital boundaries (Dyukina et al., 2025; Gleason & Greenhow, 2017; Pöysä-Tarhonen, 2025; Snart, 2017). These models reflect patterns extracted from the documents reviewed rather than definitions or examples presented in the Introduction, underscoring that blended learning ecosystems evolve dynamically across disciplines and that practical implementations often diverge from canonical theoretical frameworks. HyFlex learning refers to a dual delivery approach with maximum student choice (Nordmann et al., 2025), while phygital describes an integrated physical-digital university ecosystem (Batista-Toledo & Gavilan, 2025). Additional terminologies include blended MOOCs (BMOOCs), which combine MOOC-based online content with in-class activities (Albó et al., 2015); polysynchronous learning, integrating diverse communication channels (Design, 2016); mobile blended learning, incorporating apps alongside traditional tools (Glahn et al., 2015); and glonacal courses, merging global MOOC content with local facilitation (Norberg et al., 2015).

Blended learning environments are also described as technology-enhanced, interactive settings integrating

learning management systems (LMS), online components, and collaborative web technologies to support authentic assessments, synchronous and asynchronous learning, and co-creation experiences (Giang et al., 2025; Misra et al., 2025b; Pima et al., 2018; Zainuddin & Abdullah, 2025). Other descriptors include third generation of distance learning (Khodabandelou et al., 2016), computer-supported collaborative learning (Amhag, 2015), digital storytelling (Gogela & Ntwasa, 2015), flexible learning emphasizing education beyond institutional walls (Van Der Perre, 2015), educational informatization aligning with digital transformation agendas (W. Peng, 2017), and personal learning environments (PLEs) fostering autonomy and interactive, student-centered experiences (Boeren, 2016; Wells & Holland, 2015).

Overall, definitions converge on blended learning as an integrative, flexible, and technology-enhanced pedagogical approach that **combines learning environments, modes of participation, temporal arrangements, and technological resources in different configurations** to create interactive, engaging, and effective learning environments in higher education (Chervinska et al., 2023; Pima et al., 2018; Thai et al., 2017; Tuomainen, 2017). Its varied terminology underscores its multi-dimensional nature and alignment with the emphasis on enhancing interactivity, personalization, collaboration, and learning effectiveness through innovative educational technologies.

RQ2) What models or strategies blend face-to-face and online learning in higher education to create interactive and practical experiences?

Blended learning in higher education employs **diverse models that combine physical and virtual learning environments through different temporal, participation, and technological arrangements** to create interactive and compelling learning experiences. **In practice, these models differ in how they distribute learning across physical and virtual environments, synchronous and asynchronous interaction, student participation options, and the technological resources used to connect the different components of the course.**

The flipped classroom model is the most widely documented, reversing traditional lecture-homework sequences by placing knowledge acquisition outside class and using in-person sessions for active, critical, or problem-based learning (Adigun et al., 2024; Almeida-Bravo et al., 2025; G. Ismail et al., 2025). Applications include postgraduate statistics (Almeida-Bravo et al., 2025), English as a Second Language (ESL) supported by Artificial Intelligence (AI) tools such as ChatGPT or Duolingo (Muniandy & Selvanathan, 2025; Phanwiriyarat et al., 2025), sports education (Hasbullah, 2025), medical education that integrates design thinking to enhance

competencies (Zhang et al., 2025), and music education for motivation and problem-solving (Zhu et al., 2025). Flipped models consistently improve critical thinking (Chero-Santisteban et al., 2025; G. Ismail et al., 2025), creativity (Al-Zahrani, 2015), motivation (Tran et al., 2025; Zhou & Zhang, 2025), engagement (Dečman et al., 2025; Sailer, 2025), and self-directed learning (Z. Wang et al., 2025).

Hybrid learning models integrate the synchronous participation of in-class and remote students via videoconferencing to create fluid, collaborative spaces that blend digital and physical environments (Petäistö et al., 2025; Pöysä-Tarhonen, 2025; Ramírez-Mera et al., 2025; Vasileva & Gorbenko, 2025; Xue, 2025). [HyFlex configurations extend this flexibility by giving students greater choice over how they participate, including in-person, synchronous online, and asynchronous pathways, thereby making participation mode itself a dimension of the blend.](#) AI-driven hybrid learning enhances self-efficacy, emotional engagement, and personalization (Nong et al., 2025; Suresh Babu et al., 2025; S. Wang et al., 2025), while phygital models integrate physical and digital ecosystems for flexibility, inclusivity, and crisis resilience (Batista-Toledo & Gavilan, 2025; Sultana et al., 2025).

Project-based blended learning (PJBBL) combines face-to-face teamwork with online tasks to enhance vocational competencies (Tan et al., 2025). Problem-based learning integrated with flipped classrooms fosters critical thinking and creativity in engineering and English learning (L. Luo, 2025). MOOCs embedded in blended courses promote higher-order learning through videos and discussions, increasing participation and satisfaction when aligned with conversational frameworks (Araneta et al., 2025; Dalipi et al., 2017; Ghai et al., 2025). Models based on Small Private Online Courses (SPOCs) combine these courses with blended learning to enhance adaptability without fatigue (Jiang & Niu, 2025; Mi et al., 2025).

Comprehensive models emphasize planning, communication, technology selection, engaging content, integrated assessments, and evaluation to ensure effective blended learning design (Miranda et al., 2024). Sequential strategies combine tutorials with online components to support professional role transitions (Rossiter & Day, 2016), while dialogic learning integrates tutorials, social networks, and viva contexts to promote reflection and interaction (Simpson, 2016). The Normale Lecture Model combines lectures, group activities, asynchronous tasks, and tools like Quizizz, which is particularly effective for large classes despite challenges in instructor interaction (de Vera et al., 2025).

Other effective strategies include Community of Inquiry-based designs, collaborative coaching that enhances AI and robotics competencies (Ortiz et al., 2025), and open educational practices that build reflective, culturally

competent learning communities (Karunanayaka, 2024; Wei, 2024). Finally, LMS integration, such as MyELT, is pivotal in fostering performance and interaction in blended [English as a Foreign Language \(EFL\)](#) education (Giang et al., 2025).

Overall, these models demonstrate that interactive, effective blended learning is best achieved through the intentional integration of technology-enhanced activities with pedagogies that prioritize engagement, inclusivity, flexibility, and reflective practice (Sharma, 2025; J. M. S. Wong et al., 2025; Zainuddin & Abdullah, 2025). Collectively, the evidence indicates that despite using varied terminology and formats, blended learning models align around fostering interaction, autonomy, and structured active learning.

RQ3) To what extent do blended learning models prioritize online sessions, and how does this affect student engagement and interactivity?

Responding to RQ3, blended learning models in higher education often prioritize online sessions to enhance flexibility, self-paced learning, and content coverage; however, this prioritization has mixed impacts on student engagement and interactivity. For example, flipped classroom models place lectures online to reserve in-person time for application, problem solving, and active learning (Cervone et al., 2016; Chiang & Wang, 2015; Mortensen & Nicholson, 2015; Turro et al., 2016; Wanner & Palmer, 2016). This approach fosters more profound understanding, motivation, and critical thinking in disciplines such as mathematics, nursing, and ESL (Blair et al., 2016; Hanson, 2016; Murphy et al., 2016; Sarpparaje et al., 2018). Students appreciate the ability to revisit video lectures at their own pace, with many identifying videos as their primary learning tool (Turro et al., 2016), and report increased confidence, participation, and satisfaction (Al-Zahrani, 2015; Holroyd et al., 2025; Sarpparaje et al., 2018). However, social isolation and lack of immediate feedback during online phases remain concerns, illustrating that online-first designs can undermine the social presence component that is foundational in interactive learning environments (Dečman et al., 2025; Ng, 2018).

Several studies also reported neutral or inconsistent effects, indicating that prioritizing online sessions does not consistently improve engagement and that the effect depends strongly on design quality, interaction patterns, and contextual constraints. In flipped models that rely heavily on online sessions, engagement is mediated by the quality of the design and its integration with in-class activities. For instance, hybrid and HyFlex models maximize online flexibility by allowing synchronous or asynchronous participation, enhancing behavioral, emotional, and cognitive

engagement when interactive synchronous elements and trustful interactions are maintained (Heilporn & Lakhali, 2021; Pöysä-Tarhonen, 2025; Sultana et al., 2025).

Similarly, MOOC integration within blended courses supports independent learning and career development skills (Andone & Mihaescu, 2018a, 2018b), yet requires structured guidance to prevent disengagement. In environmental chemistry, intensive online learning packages with formative assessments correlated with higher test performance (Ramadhani et al., 2019), whereas in civil engineering, online forums were underused when assessments lacked grade incentives (W.-K. Lee et al., 2019). Studies emphasize that online-first designs enhance engagement only when accompanied by interpersonal interactions, reflective activities, and structured assessments (Bokosmaty et al., 2019; Chen et al., 2019; Steen-Utheim & Foldnes, 2018). For example, flipped classrooms incorporating reflective thinking approaches showed significant gains in pre-class engagement and participation behaviors (Chen et al., 2019).

Furthermore, sequential and partially flipped models, which combine online preparation with guided inquiry and clicker-based discussions, demonstrated improved performance and retention (Bokosmaty et al., 2019). In medical education, online modules prepared students for interactive in-person meetings, enhancing their perceptions of learning (Angadi et al., 2019; Westerlaken et al., 2019). Nevertheless, inadequate infrastructure, unequal internet access, and student resistance to adopting active online learner identities remain barriers (Charbonneau-Gowdy & Chavez, 2019; Gündüz & Akkoyunlu, 2019; Karthikeyan et al., 2024).

Overall, prioritizing online sessions in blended learning models enhances flexibility, autonomy, and cognitive preparation; however, their effectiveness in promoting engagement and interactivity critically depends on intentional integration with face-to-face components, collaborative activities, motivational scaffolds, and inclusive designs. Designing blended learning for interactive learning environments thus requires balancing online prioritization with pedagogies fostering active, affective, and social engagement to achieve meaningful, personalized, and effective higher education experiences (Peres et al., 2017; Raviolo et al., 2021; J. M. S. Wong et al., 2025).

RQ4) How do blended learning environments combine synchronous and asynchronous activities, and what impacts do these balances have on engagement and outcomes?

Responding to RQ4, blended learning environments in higher education combine synchronous and asynchronous activities through intentional design, integrating pre-class, in-class, and post-class components to

optimize engagement and learning outcomes. Flipped classroom models widely adopt asynchronous video lectures and quizzes for pre-class preparation, enabling synchronous sessions to focus on interactive, problem-based, and collaborative activities that translate preparatory knowledge into higher-order reasoning, situating students within socially rich learning exchanges (Holroyd et al., 2025; Karthikeyan et al., 2024; Sailer, 2025; Q. Wang, 2025). For example, in medical education in Zambia, asynchronous neurology lectures combined with synchronous small-group discussions have been shown to improve knowledge gains, confidence, and satisfaction compared to traditional approaches (Holroyd et al., 2025). Similarly, flipped classrooms in English language courses have been found to foster higher academic achievements and engagement (Q. Wang, 2025). However, studies note challenges such as infrastructural limitations, teacher reluctance, and student time constraints that can hinder integration (Dečman et al., 2025; Karthikeyan et al., 2024; Sailer, 2025).

Blended models also incorporate microblogging-supported hybrid courses, combining asynchronous discussions and backchanneling with synchronous live chats, thereby enhancing flexibility and preparation for in-class engagement; however, deep interactive exchanges remained limited (T. Luo et al., 2019; Shiau et al., 2018; Tosun, 2018). In medical education, concise asynchronous online preparation paired with structured synchronous active learning yielded higher test scores and reflective learning gains (X. Cheng et al., 2017; Ramnanan & Pound, 2017; Rui et al., 2017). Likewise, asynchronous pre-class preparation in web development courses enhanced motivation, confidence, and collaborative skills during synchronous activities, despite challenges in group dynamics (Sailer, 2025). Taken together, these findings show that the effectiveness of synchronous–asynchronous combinations depends on coherent design rather than the proportion of time allocated to each modality.

In engineering and mathematics education, asynchronous online learning packages rich in videos and assessments improved conceptual understanding, with performance correlating to online engagement (Cagande & Jugar, 2018; Ramadhani et al., 2019; Steen-Utheim & Foldnes, 2018). However, civil engineering students engaged less in online forums when assessments lacked grade incentives, underscoring that design quality and assessment alignment affect interactivity (W.-K. Lee et al., 2019). Inquiry-based flipped classrooms, where students prepared asynchronously and engaged synchronously in problem-solving, enhanced critical thinking and reasoning skills (Jensen-Vallin, 2017).

Blended learning **configurations influence outcomes by supporting** self-directed learning and flexible pacing. For example, histology courses integrating asynchronous video lectures with synchronous interactive quizzes

improved test scores and satisfaction (X. Cheng et al., 2017; McLean, 2018). In language education, asynchronous independent study increased synchronous opportunities for proficiency development (Huh & Yoo, 2017; Rosen et al., 2016), while flipped business statistics courses, which integrated asynchronous quizzes and synchronous exercises, significantly improved exam performance (Shinaberger, 2017).

Despite positive outcomes, some studies indicate mixed impacts. In pathology, asynchronous online modules combined with synchronous integration sessions promoted understanding and motivation (Herbert et al., 2017). However, in engineering, students valued synchronous interactions more for threshold concepts, despite being familiar with asynchronous video (Khoo et al., 2015). Additionally, heavy workloads and lack of preparation reduced the acceptance of flipped models, suggesting that optimal engagement requires workload management and scaffolding (Blair et al., 2016; Foster & Stagl, 2018).

Overall, strategic integration of asynchronous preparation with synchronous interactive sessions enhances cognitive, agentic, and emotional engagement, self-regulation, and knowledge retention, aligning with the emphasis on interactive, technology-enhanced, and learner-centered educational designs (Miranda et al., 2024; Nong et al., 2025; Peres et al., 2017; Subramaniam & Muniandy, 2019). Together, these findings show that optimal learning occurs when asynchronous preparation is purposefully connected to synchronous interaction through coherent instructional design.

RQ5) What technologies are used in higher education blended learning, and how do they support interactivity and learning outcomes?

Responding to RQ5, blended learning in higher education integrates diverse technologies to enhance interactivity and learning outcomes, and the choice of technology functions as a central design decision that shapes the nature of learner-content, learner-peer, and learner-instructor interactions. Flipped classroom models widely adopt instructor-made videos, screencasts, podcasts, and interactive quizzes to deliver content asynchronously, thereby enabling synchronous class time for problem-solving, collaborative learning, and the development of higher-order thinking skills (Johnston, 2017; Roehling, 2017; Rosen et al., 2016; Sailer, 2025). For example, in pharmacy education, dedicated e-learning lectures combined with student-led scenario-based learning improved knowledge acquisition and post-test scores. Meanwhile, flipped language learning allowed for the use of synchronous time for differentiated instruction and addressing misconceptions (Rosen et al., 2016). Learning Management Systems

(LMS) such as Moodle, MyELT, Open eClass, and MUELE facilitate flexible access to multimedia content, online assessments, peer discussions, and tutor–student interactions, supporting independent study, formative assessment, and learner engagement (Egesa et al., 2024; Giang et al., 2025; Kabassi et al., 2016; V. A. Nguyen, 2017b). LMS-based peer assessment frameworks enhance collaborative evaluation, deep understanding, and active participation (V. A. Nguyen, 2017a).

Artificial Intelligence (AI) tools are increasingly integrated, including intelligent tutoring systems, adaptive learning platforms, text mining predictive models for learning style classification with 98.75% accuracy (Nuankaew & Nuankaew, 2025; Suresh Babu et al., 2025), and AI-powered gamified applications like Duolingo, which foster vocabulary, pronunciation, and speaking skills while reducing learner anxiety (Phanwiriyarat et al., 2025). ChatGPT has been utilized as a collaborative tool in flipped ESL classrooms to provide instant feedback and enhance speaking skills (Muniandy & Selvanathan, 2025). AI-based learning analytics inform personalized feedback and adaptive pathways, with educator beliefs shaping effective implementation (McCarthy et al., 2025).

Mobile learning technologies support anytime-anywhere learning, real-time flexible interactions, and personalization through apps and [bring-your-own-device \(BYOD\)](#) strategies, advancing knowledge acquisition, collaboration, and reflective engagement (S. C. Kong & Song, 2015; M. Li, 2025; Xue, 2025; Zhao, 2025). For example, [dynamic time warping \(DTW\)](#) algorithms in Japanese flipped classes aligned module delivery with optimal circadian rhythms to enhance engagement and outcomes (M. Li, 2025). Mobile apps, such as Quizlet, improve vocabulary learning and test scores (Lander, 2015), while mobile webinars facilitate synchronous discussions, problem-solving, and group work across devices (Amhag, 2015).

Massive Open Online Courses (MOOCs) and Small Private Online Courses (SPOCs) provide asynchronous, self-paced learning that fosters reflection, syntactic fluency, and confidence in English speaking when combined with structured speaking tasks (Araneta et al., 2025; Y. Li et al., 2015; Mi et al., 2025). MOOCs integrated with flipped classrooms improve flexibility, perceived usefulness, and summative exam performance (Cutrell et al., 2015). Augmented Reality (AR) enhances cognitive presence and engagement in mechanical engineering courses, influenced by performance expectancy and hedonic motivation (Camilleri et al., 2025). Problem-based learning platforms, interactive digital storytelling, and immersive simulations further foster critical thinking, creativity, and professional competencies across disciplines (Gogela & Ntwasa, 2015; Köttgen et al., 2016; L. Luo, 2025).

Overall, these technological integrations support learner–content, learner–learner, and learner–teacher

interactivity, promoting self-regulated, collaborative, and personalized learning experiences. Effective implementation requires early planning, clear communication, technological scaffolding, and active learning designs (Miranda et al., 2024; Misra et al., 2025b), [aligning with the goal of creating engaging, adaptive, ethical, and practical educational experiences in higher education.](#)

RQ6) What student learning, engagement, and satisfaction outcomes are reported in blended learning, and how do these inform effective interactive design?

Responding to RQ6, blended learning in higher education demonstrates consistently positive outcomes in student learning, engagement, and satisfaction, providing clear insights for effective interactive design. Learning outcomes improve through flipped classrooms, where critical thinking, comprehension, and problem-solving skills are enhanced across disciplines such as English reading (Tran et al., 2025), music theory (Zhu et al., 2025), linear algebra (Murphy et al., 2016), and chemistry (Hibbard et al., 2016), with reported gains in median exam scores and conceptual understanding. Project-based blended learning enhances vocational competencies and team spirit (Tan et al., 2025), while industry-based authentic assessment develops professional skills in [Information Technology \(IT\)](#) education (Sandanayake & Perera, 2025). Peer evaluation fosters critical thinking and reflective consciousness (Q. Wu & Yang, 2025), and flipped approaches that integrate AI tools, such as intelligent tutoring systems, promote personalization and tailored learning pathways (Suresh Babu et al., 2025).

Engagement outcomes reveal [higher participation, motivation, and confidence](#) when asynchronous pre-class videos are paired with synchronous interactive sessions. Weekly online participation, collaborative group activities, and forum interactions are strong predictors of improved exam performance in social sciences and language courses (Dečman et al., 2025; L. Luo, 2025). Flipped classrooms, combining design thinking and collaborative coaching, cultivate self-awareness, teamwork, and technical skills, as evidenced by international robotics competition success (Ortiz et al., 2025; Zhang et al., 2025). Engagement is also driven by perceived autonomy, relatedness, and competence, without gender differences (Y. Yang & Tan, 2025). Meanwhile, factors such as teacher presence, dialogic exchanges, and structured in-class activities remain essential for translating engagement into performance gains because they anchor online preparation within coherent pedagogical sequences that reinforce motivation and conceptual understanding (Hotle & Garrow, 2016; Peter et al., 2017; Simpson, 2016).

Satisfaction outcomes indicate students value blended learning for flexibility, personalized experiences, and

enhanced digital literacy (Gogela & Ntwasa, 2015; Gupta & Raj, 2024). For example, nursing and medical education students report improved confidence and knowledge retention with flipped learning (Holroyd et al., 2025; D. Wu et al., 2025). However, satisfaction varies depending on design alignment; web-based lectures are preferred as preparation for practical activities rather than as course extensions (Montrieux et al., 2015). Large, blended classes face challenges such as limited instructor interaction, technological issues, and stress (de Vera et al., 2025), highlighting the need for robust institutional support, continuous faculty training, and reliable infrastructure (Y. Peng et al., 2025). Blended learning designs integrating mobile apps, LMS platforms, and AI tools foster interactivity and satisfaction by supporting flexible, self-directed, and collaborative learning (Giang et al., 2025; S. C. Kong & Song, 2015; Nuankaew & Nuankaew, 2025).

Overall, these findings inform effective interactive design by emphasizing: (1) structured asynchronous preparation coupled with active synchronous sessions; (2) integration of AI, LMS, and mobile technologies to personalize learning; (3) incorporation of peer evaluation, problem-based, and authentic assessments to build critical thinking and professional competencies; (4) teacher presence, feedback, and adaptive teaching to maintain engagement; and (5) institutional support ensuring technological readiness and faculty development (Altowairiki, 2025; Miranda et al., 2024; Y. Peng et al., 2025; J. M. S. Wong et al., 2025). Designing blended learning that nurtures autonomy, competence, and relatedness, while aligning pedagogical and technological elements with learner needs, maximizes learning effectiveness, engagement, and satisfaction in interactive higher education environments. Collectively, these results highlight that technology and pedagogy reinforce each other most effectively when aligned with autonomy, feedback, and structured engagement opportunities. At the same time, some studies reported outcomes that were comparable to those of traditional instruction, suggesting that the benefits of blended learning are modest and strongly mediated by design quality, contextual conditions, and student readiness.

RQ7) What barriers do educators face in implementing blended learning, and how do these affect the adoption of interactive learning practices?

Responding to RQ7, **educators** face multifaceted barriers when implementing blended learning, **and these barriers directly affect the adoption** of interactive learning practices in higher education. Infrastructural barriers, such as inadequate internet connectivity, inflexible physical spaces, and poor technological infrastructure, restrict

flexibility and hinder the integration of blended models, as seen in medical universities and STEM disciplines (Abbasi et al., 2025; Hagerup et al., 2017; J. M. S. Wong et al., 2025). Time constraints consistently emerge; teachers report insufficient time for preparation, video production, and redesigning course materials, which limits their ability to implement flipped classrooms and interactive pedagogies effectively (Antonova et al., 2017; Liu et al., 2019).

Most of the studies addressing these challenges originate from higher-income institutions, suggesting that barriers in lower-resource contexts may be underrepresented in the available evidence. These constraints were more prominent in resource-limited institutions, whereas educators in higher-resourced contexts emphasized challenges related to course redesign, workload recognition, and coordination. The lack of institutional support compounds these issues, with limited training, the absence of strategic frameworks, and inadequate recognition of workload for online teaching tasks reducing motivation for blended adoption (Egesa et al., 2024; Garone et al., 2025; Porter & Graham, 2016).

Pedagogical barriers include low self-efficacy in using digital tools, limited technological pedagogical content knowledge (TPACK), and lack of experience in designing learner-centered, interactive activities (Alammary et al., 2015; Ali, 2024a; Shu, 2015). Faculty often express uncertainty about course organization in flipped learning, resistance to changing traditional practices, and discomfort with shifting pedagogic identities towards technology integration (Charbonneau-Gowdy & Frenzel, 2018; Jong, 2019; Sanagavarapu, 2018). Even when teachers value blended learning, cultural resistance and insufficient adaptation time limit its interactive potential (Chellapan & Van der Meer, 2015; G. K. W. Wong & Cheung, 2015). Furthermore, student-related barriers, such as low readiness for self-directed learning, concentration difficulties in online environments, and digital divides, pose challenges to educators' efforts to design equitable and engaging interactive experiences (Basitere & Ivala, 2017; J. M. S. Wong et al., 2025).

Institutional and organizational barriers remain significant. The absence of formal policies, limited project management, a lack of peer support, and poor alignment between institutional culture and blended initiatives lead to fragmented implementation, thereby undermining the systematic integration of interactive learning (Adekola et al., 2017; Bilal et al., 2022; Paul, 2019). For example, inadequate student preparation in blended models and misaligned expectations in flipped classrooms can reduce engagement and learning outcomes (Bocconi & Trentin, 2014; Jesurasa et al., 2017). Professional development gaps, where educators lack training in blended design and

facilitation, limit their ability to implement assessments, discussions, and collaborative activities that are essential for interactivity (Chaberek-Karwacka & Malinowska, 2015; Nkhukhu-Orlando, 2015). Workload concerns further demotivate faculty, as time spent on the LMS and blended design is often not formally recognized, thereby diminishing the adoption incentives (Antonova et al., 2017; Egesa et al., 2024).

Finally, contextual and sociocultural barriers, such as class size, disciplinary traditions, economic inequalities, and privacy concerns, complicate adoption (I. N. Y. Cheng et al., 2016; T. N. Nguyen et al., 2018). For example, flipped classrooms in nursing faced resistance from colleagues and students due to institutional inertia (Bernard & Ghaffari, 2019), while integrating MOOCs required educators and learners to acquire new conceptual and technological skills (Andone & Vasiu, 2016). These interconnected barriers reveal that effective adoption of blended learning as an interactive paradigm requires holistic strategies, including institutional investment in infrastructure and training, recognition of faculty workload, culturally responsive pedagogical redesigns, and supportive policies fostering teacher confidence, student readiness, and technological integration to transform higher education into engaging, personalized, and effective interactive learning environments.

Discussion

This review reveals several cross-cutting patterns across the seven research questions. Definitions and models remain fragmented, yet the functional emphasis on flexibility, interaction, and self-regulated learning is consistent across disciplines. Technologies support engagement most effectively when aligned with structured pedagogical designs, while barriers persist at technological, institutional, and pedagogical levels. These cross-cutting patterns are derived from studies spanning diverse disciplines, institutional contexts, and research designs, so the conclusions should be read as broadly indicative rather than narrowly generalizable.

These patterns align closely with the theoretical frameworks guiding this review. Constructivism explains the benefits observed in active and collaborative designs; the Community of Inquiry framework clarifies variations in engagement, satisfaction, and presence; and the TPACK model accounts for misalignments underlying many implementation challenges. Together, these perspectives provide a coherent interpretive lens for understanding the empirical evidence synthesized in the Results. [More specifically, constructivism helps interpret the active and collaborative models identified in RQ2–RQ4; CoI provides a lens for the interaction, engagement, and satisfaction patterns reported in RQ3, RQ4, and RQ6; and TPACK helps explain the relationships between technological](#)

integration and instructional design in RQ5 and the technological–pedagogical barriers identified in RQ7.

Practical implications

This section focuses on actions that can be implemented at the course, instructional design, and teaching practice levels. To ensure coherence with the conceptual framework used in the Introduction, the practical implications are organized in alignment with the five key dimensions addressed in this review: (1) definitions of blended learning, (2) models and temporal structures, (3) technologies, (4) learning outcomes, and (5) implementation barriers.

The findings of this review inform higher education practice as follows:

- Design blended learning through intentional combinations of physical and virtual environments, temporal arrangements, participation options, and technological resources to enhance flexibility, quality, engagement, motivation, critical thinking, and higher-order skills.
- Adopt flipped classroom models using online preparation and active in-class learning to foster deep understanding and self-directed learning.
- Integrate learning management systems to facilitate multimedia access, assessments, discussions, peer feedback, and collaborative learning experiences.
- Utilize AI tools like ChatGPT and Duolingo for personalized, adaptive, and effective language and skill development.
- Design blended courses with intentional alignment of asynchronous and synchronous components to optimize participation and learning outcomes.
- Invest in robust infrastructure, flexible spaces, faculty workload policies, and professional development to support the adoption of blended teaching.
- Embed authentic, project-based, and industry-linked assessments to strengthen students' professional competencies and real-world readiness.
- Promote hybrid and HyFlex models to enhance inclusivity, flexibility, crisis resilience, and equitable participation across learning environments.

Policy implications

To maintain consistency with the framework outlined in the Introduction and used to structure the results, the policy implications are presented according to the exact five dimensions: (1) definitions of blended learning, (2) models and temporal structures, (3) technologies, (4) learning outcomes, and (5) implementation barriers. From the analysis of the research questions, [the following policy implications were identified](#):

- Prioritize investments in robust infrastructure, reliable internet, and flexible learning spaces to enable the adoption of equitable, scalable, and effective blended learning across disciplines and institutional contexts.
- Institutionalize blended learning as a core instructional mode through strategic frameworks, formal policies, and systematic implementation at the activity, course, and program levels, beyond emergency remote teaching.
- Adopt policies that support flipped classrooms, incorporating online knowledge acquisition and active, problem-based, collaborative in-class learning to enhance critical thinking, motivation, engagement, and self-directed learning skills.
- Develop policies that recognize faculty workload and allocate time for preparation, video production, and course redesign, alongside continuous professional development, to build blended teaching expertise and confidence.
- Mandate integration of LMS platforms to facilitate multimedia content access, online assessments, peer discussions, collaborative learning, and learner-centered designs, fostering autonomy, engagement, and digital competencies.
- Promote policies that integrate AI tools, mobile learning, and BYOD strategies to enable personalized, adaptive, and anytime-anywhere learning, aligned with digital transformation and educational informatization agendas.
- Encourage the integration of project-based, problem-based, and authentic assessments within blended courses to develop vocational competencies, professional skills, critical thinking, creativity, and reflective consciousness.
- Embed hybrid and HyFlex models into institutional policies to foster inclusivity, flexibility, crisis resilience, and [meaningful student choice among in-person, synchronous online, and asynchronous](#)

participation pathways.

Limitations of the Study

This study has several limitations that should be considered when interpreting its findings. First, the review relies exclusively on Scopus. Although this database provides broad multidisciplinary coverage and enables the use of a consistent bibliographic structure throughout the retrieval and analysis process, relevant studies indexed only in other databases or disseminated through regional or less widely indexed outlets may not have been captured. Consequently, the review should be interpreted as a synthesis of the Scopus-indexed literature on blended learning in higher education rather than as an exhaustive representation of all available research.

Second, the study intentionally prioritized the 100 most recent documents considered relevant to each research question. This strategy was adopted to foreground contemporary developments in a field characterized by rapid changes in educational technologies, delivery models, and post-pandemic practices. However, it may have excluded earlier studies that remain influential or theoretically important. In addition, because eligibility was assessed separately for each research question, the sets of documents contributing to individual findings are not necessarily identical.

Third, retrieval remains dependent on the terminology used to describe blended learning. Although the search strategy was iteratively refined to incorporate alternative terms and related formulations, the field uses overlapping concepts such as blended, hybrid, flipped, flexible, and technology-enhanced learning. Studies using terminology not represented in the search strategy may therefore have been overlooked.

Finally, the scoping-review design emphasizes mapping and synthesizing the breadth, characteristics, and recurring patterns of the literature rather than estimating intervention effects or comparatively evaluating the methodological rigor of individual studies. No formal quality appraisal or risk-of-bias assessment was therefore used to weight the evidence. The reported findings should consequently be interpreted as patterns across a heterogeneous body of literature rather than as precise estimates of the effectiveness of particular blended-learning models, technologies, or instructional strategies.

Conclusions

To align this section with the organizational framework used in the Introduction and throughout the results, the conclusions synthesize the main findings following the same five dimensions, thereby consolidating contemporary evidence into an integrated structure that is rarely presented cohesively in prior reviews: (1) definitions of blended learning, (2) models and temporal structures, (3) technologies, (4) learning outcomes, and (5) implementation barriers.

In addition, the review directly addresses each of the seven research questions. RQ1 clarifies how blended learning is defined and described across the literature; RQ2, RQ3, and RQ4 synthesize the pedagogical models, temporal structures, and balances of synchronous and asynchronous components; RQ5 and RQ6 integrate evidence on technological support and learning outcomes; and RQ7 identifies the institutional, pedagogical, and technological barriers shaping implementation.

Blended learning in higher education integrates [physical and virtual learning environments](#), [temporal arrangements](#), [participation modes](#), and [technological resources in different configurations](#) to create flexible, personalized, and interactive experiences, functioning as a strategic response to the increasing need for adaptable, technology-mediated learning ecosystems. These findings reinforce the theoretical expectations outlined earlier. Constructivism explains the strong performance of designs that blend preparation with collaborative inquiry; CoI clarifies how interaction patterns influence engagement, satisfaction, and presence; and TPACK accounts for persistent adoption challenges related to technological and pedagogical misalignment. Thus, the empirical evidence supports and extends these theoretical predictions.

Models such as flipped classrooms, hybrid and HyFlex designs, MOOC-integrated approaches, AI-powered adaptive learning, and project-based frameworks demonstrate consistent benefits in motivation, self-regulation, and higher-order skills when technology is intentionally aligned with pedagogy. However, implementation faces significant barriers. Educators report infrastructural limitations, inadequate technological resources, and time constraints that restrict effective design. Negative beliefs, low self-efficacy, and cultural resistance further limit adoption, while organizational challenges, including a lack of strategic frameworks, insufficient institutional support, and the absence of coordinated policies, constrain scalability and sustainability. Student-related barriers, including digital divides, inadequate preparation for self-directed learning, and limited social presence in online components, hinder engagement and inclusivity. Addressing these challenges requires holistic strategies that

integrate faculty training, robust infrastructure, supportive institutional policies, and culturally responsive practices to realize the transformative potential of blended learning.

Future initiatives should prioritize adaptive technologies, inclusive design, and authentic assessments to create engaging, ethical, and compelling interactive environments that build lifelong learning skills and the digital competencies essential to contemporary higher education. Longitudinal research is also needed to examine sustained learning trajectories, discipline-specific impacts, and the long-term effectiveness of emerging technologies such as AI-mediated personalization. Overall, blended learning presents a promising pathway to reimagine higher education through interactive, learner-centered, and technology-enhanced pedagogies that align with evolving educational and societal needs.

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The authors ...

References

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